

Image Generation using (β -)VAE, VQ-VAE and WGAN

Assignment Overview

In this assignment, you will implement and compare generative modeling approaches for images:

- (β -)Variational Autoencoder (β -VAE)
- Vector Quantized VAE (VQ-VAE) with PixelCNN prior for latent sampling
- Wesserstein Generative Adversarial Network (WGAN) and WGAN-GP

The goal is to understand how different techniques influence **reconstruction quality**, and **sample realism** in images. Use provisioned WILP lab infrastructure (mandatory)!

Dataset

Use **CIFAR 10** dataset.

Preprocessing

- Normalize pixel values to [0,1]

Part A:

(β -)VAE [2 Marks]

Use $\beta \in \{1, 2, 4, 10\}$

Tasks

- Train models with different β values
- Compare reconstruction quality (PSNR) w.r.t. epochs for different β values
- Compare FID scores for different β values

VQ-VAE with PixelCNN Prior [3 Marks]

Use Codebook size K (512, 256, 128)

Tasks

- Train VQ-VAE and report reconstruction quality
- Extract discrete latent maps
- Train PixelCNN over latents
- Sample new latent codes using PixelCNN

- • Generate images from sampled latents
- • Compare FID scores, reconstruction quality (PSNR) and codebook quality for different K

Part B: WGAN and WGAN-GP [3 Marks]

Tasks

- • Train WGAN and WGAN-GP
- • Generate random samples
- • Compute FID scores

Part C: Comparative Analysis [2 Mark]

Include 100 visual examples for VAE ($\beta=1$), VQ-VAE ($K=256$), WGAN and WGAN-GP and comment on perceptual quality.

Compare evaluation scores for (β -)VAE, VQ-VAE, WGAN and WGAN-GP

Deliverables

- • **ipynb** notebooks (PyTorch / TensorFlow) for Part A, and Part B
- • Corresponding html files
- • Pdf file for Part C including comparative discussion (reconstruction error, sampling quality, speed)

Marks will be deducted for unsatisfactory values of evaluation metrics.